

CHAPTER

14

Time-Dependent Perturbation Theory

In Chapters 10 and 12, we studied *time-independent* perturbation theory and found that changes in the Hamiltonian lead to changes in the energy eigenstates of a system. In those examples, the Hamiltonian was not time dependent, so the perturbed energy levels were still stationary states of the system. Now we turn to the problem of understanding how a system responds to changes in the Hamiltonian that are a function of time. We will find that the new perturbed energy states are no longer stationary states and that changes or **transitions** between states can occur. In Chapter 3, we solved the time-dependent case exactly for a sinusoidal perturbation of the two-level spin system. We found that spin flips or transitions between spin up and down states occur when the frequency of the time dependence is close to the Bohr frequency characterizing the energy splitting of the two states. This resonance condition is also an important idea in this chapter.

The transitions between energy states that arise from a time-dependent Hamiltonian play a major role in experimental studies of quantum mechanical systems. We have referred many times to spectroscopic experiments that provide evidence of the energies of quantum systems. These spectroscopic experiments rely on the interaction between the oscillating electromagnetic fields of laser beams and atoms or molecules that respond to these time-dependent fields. The examples in this chapter will help us better understand these light-matter interactions.

14.1 ■ TRANSITION PROBABILITY

The typical experiment that we wish to model with time-dependent perturbation theory is the following: we start with a system in a particular initial quantum state $|i\rangle$, we turn on a perturbing Hamiltonian $H'(t)$ at time $t = 0$, and then we measure the probability that the system is in a new final quantum state $|f\rangle$ at a later time. For example, a hydrogen atom in its ground state $|1s\rangle$ is perturbed by an incident laser beam, and we wish to know the probability of the atom making a transition to the $|3p\rangle$ excited state. The Hamiltonian is assumed to be H_0 before the perturbation, and as in time-independent perturbation theory, we assume that we know the solutions to the unperturbed energy eigenvalue equation:

$$H_0|n\rangle = E_n|n\rangle. \quad (14.1)$$

In the hydrogen example, H_0 is the hydrogen atom Hamiltonian and E_n and $|n\rangle$ (shorthand for $|n\ell m\rangle$) are the eigenenergies and eigenstates we solved for in Chapter 8.

The Schrödinger equation that governs the time evolution of a quantum system is

$$H|\psi\rangle = i\hbar \frac{d}{dt}|\psi\rangle. \quad (14.2)$$

The full Hamiltonian

$$H = H_0 + H'(t) \quad (14.3)$$

is now time dependent, so we cannot follow the standard recipe we developed in Chapter 3 for determining the time evolution of the quantum state vector. In principle, we have to rediagonalize the Hamiltonian and find the new energy eigenstates, and then do that each time the Hamiltonian changes. Because the Hamiltonian is continuously changing, that is nearly impossible to do.

Rather, we take an approach that is similar to that taken in time-independent perturbation theory: we assume the perturbation is small enough that the zeroth-order energy eigenstates are a good approximation for starting the solution. But now we are more interested in solving the Schrödinger equation than in solving the energy eigenvalue equation. We are not so interested in how the perturbation changes the energies of the states; rather, we want to find how the perturbation changes the time evolution of the system. We use the original energy basis for expanding general states of the system, even though these states may not be energy eigenstates of the perturbed system.

Using the zeroth-order energy basis, the initial state of the system, before the perturbation is turned on, is

$$|\psi(t=0)\rangle = \sum_n c_n |n\rangle. \quad (14.4)$$

We know from the Schrödinger recipe of Chapter 3 that the time evolution of this initial state without any perturbation would be

$$|\psi_{H=0}(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle, \quad (14.5)$$

where each term acquires a time-dependent phase evolution factor dependent on the energy of that term. The application of the perturbation $H'(t)$ gives rise to new energy eigenstates and hence new time evolution phase factors. However, if the perturbation is small, then we expect that the new solution will be close to the zeroth-order solution of Eq. (14.5). Hence, we assume that we can modify the zeroth-order solution by including another factor that reflects the additional time dependence caused by the perturbation. We do this by allowing the expansion coefficients to be time dependent:

$$|\psi_{H \neq 0}(t)\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle. \quad (14.6)$$

Now our task is to determine how the coefficients $c_n(t)$ depend on time, with the obvious restriction that they equal their original values $c_n(0)$ at $t = 0$. Substituting Eq. (14.6) for the time evolved state into the Schrödinger equation (14.2), we find

$$\begin{aligned} (H_0 + H'(t))|\psi(t)\rangle &= i\hbar \frac{d}{dt}|\psi(t)\rangle \\ (H_0 + H'(t)) \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle &= i\hbar \frac{d}{dt} \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle, \end{aligned} \quad (14.7)$$

and using the zeroth-order energy eigenvalue equation (14.1) to cancel some terms yields

$$\begin{aligned} \sum_n \left[E_n c_n(t) e^{-iE_n t/\hbar} |n\rangle + H'(t) c_n(t) e^{-iE_n t/\hbar} |n\rangle \right] &= i\hbar \sum_n \left[\frac{dc_n(t)}{dt} e^{-iE_n t/\hbar} |n\rangle - \frac{E_n}{\hbar} c_n(t) e^{-iE_n t/\hbar} |n\rangle \right] \\ \sum_n H'(t) c_n(t) e^{-iE_n t/\hbar} |n\rangle &= i\hbar \sum_n \frac{dc_n(t)}{dt} e^{-iE_n t/\hbar} |n\rangle. \end{aligned} \quad (14.8)$$

To simplify this differential equation, we isolate one coefficient in the sum on the right side by projecting the whole equation onto a particular energy state, say $|k\rangle$, and use orthogonality to find

$$\begin{aligned} \langle k | \sum_n H'(t) c_n(t) e^{-iE_n t/\hbar} |n\rangle &= \langle k | i\hbar \sum_n \frac{dc_n(t)}{dt} e^{-iE_n t/\hbar} |n\rangle \\ \sum_n c_n(t) e^{-iE_n t/\hbar} \langle k | H'(t) |n\rangle &= i\hbar \frac{dc_k(t)}{dt} e^{-iE_k t/\hbar}. \end{aligned} \quad (14.9)$$

Rearranging terms yields a differential equation

$$i\hbar \frac{dc_k(t)}{dt} = \sum_n c_n(t) e^{i(E_k - E_n)t/\hbar} \langle k | H'(t) |n\rangle \quad (14.10)$$

for each coefficient c_k of the expansion. This result is still exact, but it gives us a set of coupled differential equations that is difficult to solve. We seek a perturbative solution by using an iterative approach. We expand the coefficient c_n in a perturbative series

$$c_n = c_n^{(0)} + c_n^{(1)} + c_n^{(2)} + \dots, \quad (14.11)$$

where the superscript denotes the order of the perturbation. The right side of Eq. (14.10) already has one order of the perturbation in $H'(t)$, so when we equate the two sides of the equation to the same order of the perturbation, we end up with the order of c_n on the right side being one less than the order on the left side (Problem 14.1). The zeroth-order term of Eq. (14.10) is

$$i\hbar \frac{dc_k^{(0)}(t)}{dt} = 0. \quad (14.12)$$

This says that the coefficients c_n have no time dependence when there is no perturbation, which is consistent with Eq. (14.5) where all the Schrödinger evolution time dependence is already specified. The first-order term of Eq. (14.10) is

$$i\hbar \frac{dc_k^{(1)}(t)}{dt} = \sum_n c_n^{(0)}(t) e^{i(E_k - E_n)t/\hbar} \langle k | H'(t) |n\rangle. \quad (14.13)$$

We can continue in this manner to all orders if we wish, but we will not go beyond the first-order solution. To collapse the sum on the right side of Eq. (14.13), we make the assumption mentioned above that the system starts in one particular eigenstate $|i\rangle$ of the zeroth-order Hamiltonian, that is $|\psi(0)\rangle = |i\rangle$. Thus the initial coefficients obey:

$$c_n(0) = \delta_{ni}. \quad (14.14)$$

In zeroth-order there is no time dependence, according to Eq. (14.12), so we obtain

$$c_n^{(0)}(t) = \delta_{ni}. \quad (14.15)$$

Substituting Eq. (14.15) into Eq. (14.13) collapses the sum to just one term and yields the first-order differential equation for the coefficient $c_k(t)$:

$$i\hbar \frac{dc_k^{(1)}(t)}{dt} = e^{i(E_k - E_i)t/\hbar} \langle k | H'(t) | i \rangle. \quad (14.16)$$

We solve Eq. (14.16) by integrating directly to give

$$c_k(t) = \frac{1}{i\hbar} \int_0^t \langle k | H'(t') | i \rangle e^{i(E_k - E_i)t'/\hbar} dt'. \quad (14.17)$$

We have dropped the superscript on $c_k(t)$ because $c_k^{(0)}(t) = 0$ for $k \neq i$ and we will not solve for higher-order terms, so Eq. (14.17) gives us the complete coefficient to our desired order. Equation (14.17) tells us how the expansion coefficient $c_k(t)$ for the energy eigenstate $|k\rangle$ evolves with time subject to the perturbation $H'(t)$, given that the system started in the state $|i\rangle$. Equation (14.17) has the familiar form of a Fourier transform, so we interpret the result as the Fourier coefficient (in frequency space) of the perturbation $H'(t)$ at the Bohr frequency

$$\omega_{ki} = \frac{E_k - E_i}{\hbar}. \quad (14.18)$$

If the perturbation $H'(t)$ has an appreciable component at the Bohr frequency ω_{ki} associated with the energy difference between the initial state $|i\rangle$ and some other state $|k\rangle$, then the probability of the system making a transition from the initial state $|i\rangle$ to the state $|k\rangle$ is large.

To find the probability that the system is measured to be in a particular final state $|f\rangle$ at a later time, we project the time-evolved state Eq. (14.6) onto the final state

$$\begin{aligned} \mathcal{P}_{i \rightarrow f}(t) &= |\langle f | \psi(t) \rangle|^2 \\ &= \left| \langle f | \sum_n c_n(t) e^{-iE_n t/\hbar} | n \rangle \right|^2 \\ &= |c_f(t)|^2 \end{aligned} \quad (14.19)$$

and substitute Eq. (14.17) to obtain

$$\mathcal{P}_{i \rightarrow f}(t) = \frac{1}{\hbar^2} \left| \int_0^t \langle f | H'(t') | i \rangle e^{i(E_f - E_i)t'/\hbar} dt' \right|^2. \quad (14.20)$$

Of course, to actually do the integral and find the probability, we need to know the form of the perturbation $H'(t)$.

Example 14.1: Constant perturbation The simplest example of a time-dependent perturbation is one that is turned on at $t = 0$ and then turned off at a later time, but that is constant during the time it is on. The integral in Eq. (14.17) to find the coefficient $c_f(t)$ of the final state is straightforward:

$$\begin{aligned} c_f(t) &= \frac{1}{i\hbar} \langle f | H' | i \rangle \int_0^t e^{i\omega_{fi}t'} dt' \\ &= \frac{1}{i\hbar} \langle f | H' | i \rangle \frac{e^{i\omega_{fi}t} - 1}{i\omega_{fi}} \\ &= \frac{1}{i\hbar} \langle f | H' | i \rangle e^{i\omega_{fi}t/2} \frac{e^{i\omega_{fi}t/2} - e^{-i\omega_{fi}t/2}}{i\omega_{fi}} \\ &= \frac{2}{i\hbar} \langle f | H' | i \rangle e^{i\omega_{fi}t/2} \frac{\sin(\omega_{fi}t/2)}{\omega_{fi}}. \end{aligned} \quad (14.21)$$

The probability that the system is measured in the final state $|f\rangle$ is

$$\mathcal{P}_{i \rightarrow f}(t) = |c_f(t)|^2 = \frac{4|\langle f | H' | i \rangle|^2}{\hbar^2 \omega_{fi}^2} \sin^2(\omega_{fi}t/2). \quad (14.22)$$

This agrees with the Rabi formula found in Chapter 3 for the probability of a spin flip caused by a small perturbing constant magnetic field [Eq. (3.63)]. Equation (14.22) tells us that to make the transition from the state $|i\rangle$ to the state $|f\rangle$, there are two essential requirements: (1) the matrix element $\langle f | H' | i \rangle$ that determines whether the perturbation connects the two levels must be nonzero, and (2) to get appreciable probability, there must be frequency components in the time-dependent Hamiltonian that include the Bohr frequency ω_{fi} for the transition. The first requirement is related to the selection rules that we have mentioned previously and that we discuss more fully in Section 14.4. The second requirement is the resonance condition inherent in the Fourier integral of Eq. (14.17). Even though the perturbation $H'(t)$ in this example is constant during its application, there is a frequency component at the Bohr frequency ω_{fi} arising from the off-on-off time dependence.

Example 14.2: Gaussian perturbation Now let's add some more interesting time dependence by assuming that a perturbation is turned on and then turned off with a Gaussian time dependence as shown in Fig. 14.1. The form of the perturbation is

$$H'(t) = V_0 e^{-t^2/\tau^2}, \quad (14.23)$$

where τ is the characteristic time constant of the perturbation. This perturbation is peaked at $t = 0$ and becomes minimal a few time constants away from that. It differs mathematically from the situation we had above where the perturbation started at $t = 0$. We accommodate this change by shifting the starting time of the integral in Eq. (14.17). The major contribution to the integral comes from times that are a few time constants before and after the peak at $t = 0$, but mathematically it

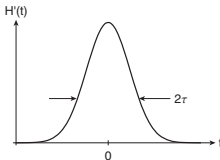


FIGURE 14.1 Gaussian time dependence of perturbation.

is simpler to integrate between $\pm\infty$. The coefficient $c_f(\infty)$ after the perturbation has been applied is therefore

$$\begin{aligned} c_f(\infty) &= \frac{1}{i\hbar} \int_{-\infty}^{\infty} \langle f | H'(t') | i \rangle e^{i(E_f - E_i)t'/\hbar} dt' \\ &= \frac{1}{i\hbar} \int_{-\infty}^{\infty} \langle f | V_0 | i \rangle e^{-t'^2/\tau^2} e^{i\omega_{fi}t'} dt'. \end{aligned} \quad (14.24)$$

The $\sin(\omega_{fi}t')$ part of the complex exponential $e^{i\omega_{fi}t'}$ is odd with respect to $t' = 0$, so that part of the integral is zero, giving

$$\begin{aligned} c_f(\infty) &= \frac{1}{i\hbar} \langle f | V_0 | i \rangle \int_{-\infty}^{\infty} e^{-t'^2/\tau^2} \cos(\omega_{fi}t') dt' \\ &= \frac{1}{i\hbar} \langle f | V_0 | i \rangle \sqrt{\pi\tau} e^{-\omega_{fi}^2\tau^2/4}. \end{aligned} \quad (14.25)$$

The probability that after the perturbation the system is measured in the final state $|f\rangle$ is

$$\mathcal{P}_{i \rightarrow f} = |c_f(\infty)|^2 = \frac{\pi\tau^2}{\hbar^2} |\langle f | V_0 | i \rangle|^2 e^{-\omega_{fi}^2\tau^2/2}. \quad (14.26)$$

This result tells us that to have appreciable probability for the transition from the state $|i\rangle$ to the state $|f\rangle$, the time constant τ must be of order $1/\omega_{fi}$, so that there are frequency components in the time-dependent Hamiltonian that include the Bohr frequency ω_{fi} for the transition.

An important lesson from this example concerns a perturbation that is turned on and off very slowly (i.e., the time constant is very long compared with other times relevant to the system). As the time constant τ becomes large enough that the product $\omega_{fi}\tau$ approaches infinity, the probability $\mathcal{P}_{i \rightarrow f}$ in Eq. (14.26) approaches zero, meaning that the system does not change states. This is an example of the adiabatic theorem in quantum mechanics.

14.2 ■ HARMONIC PERTURBATION

The previous examples have illustrated the importance of frequency components that match the Bohr frequency of the transition. Frequency components in the time dependence of the Hamiltonian that are

far from the Bohr frequency of a particular transition do not produce appreciable probability for that transition. Hence, the most efficient way to make a transition is to impose a sinusoidal perturbation *at the transition frequency*. The study of such resonant interactions is the most important example of time-dependent perturbation theory.

At $t = 0$ we turn on a time-dependent perturbation Hamiltonian that has separate space and time parts:

$$\begin{aligned} H'(t) &= 2V(\vec{r})\cos\omega t \\ &= V(\vec{r})(e^{i\omega t} + e^{-i\omega t}). \end{aligned} \quad (14.27)$$

There are different conventions for including the factor of 2 in Eq. (14.27) or not; without it, one needs a factor of 1/2 for each complex exponential. Substituting this harmonic perturbation into Eq. (14.17) yields the probability amplitude for making a transition from an initial state $|i\rangle$ to a final state $|f\rangle$:

$$\begin{aligned} c_f(t) &= \frac{1}{i\hbar} \int_0^t \langle f | V(\vec{r}) (e^{i\omega t'} + e^{-i\omega t'}) | i \rangle e^{i(E_f - E_i)t'/\hbar} dt' \\ &= \frac{1}{i\hbar} \langle f | V | i \rangle \int_0^t \left[e^{i(\omega_f + \omega)t'} + e^{i(\omega_f - \omega)t'} \right] dt' \\ &= \frac{1}{i\hbar} \langle f | V | i \rangle \left[\frac{e^{i(\omega_f + \omega)t} - 1}{i(\omega_f + \omega)} + \frac{e^{i(\omega_f - \omega)t} - 1}{i(\omega_f - \omega)} \right] \\ &= \frac{1}{i\hbar} \langle f | V | i \rangle \left[e^{i(\omega_f + \omega)t/2} \frac{\sin \frac{\omega_f + \omega}{2} t}{\frac{\omega_f + \omega}{2}} + e^{i(\omega_f - \omega)t/2} \frac{\sin \frac{\omega_f - \omega}{2} t}{\frac{\omega_f - \omega}{2}} \right]. \end{aligned} \quad (14.28)$$

To find the probability, we square this amplitude, which leads to cross terms and a complicated expression. This is what we have to do if the two terms inside the square brackets are of comparable size, which happens if the frequency is far from a resonance. However, if the resonance condition is satisfied or nearly satisfied, then one of the two terms inside the square brackets dominates because the denominator approaches zero. Which term dominates depends on the sign of the energy difference $E_f - E_i = \hbar\omega_{fi}$.

1) If the initial state is lower in energy than the final state, then the energy difference $E_f - E_i = \hbar\omega_{fi}$ is positive and the second term in Eq. (14.28) is large for an excitation frequency that matches the Bohr frequency: $\omega = \omega_{fi}$. In this case, the dominant probability amplitude is for the transition from a lower state to an upper state, which we call **absorption** [see Fig. 14.2(a)]. The system absorbs energy from the external perturbation.

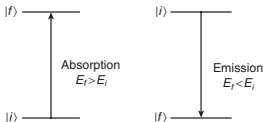


FIGURE 14.2 (a) Absorption and (b) emission processes.

2) If the initial state is higher than the final state, then the energy difference $E_f - E_i = \hbar\omega_{fi}$ is negative and the first term in Eq. (14.28) is large for an excitation frequency that matches the Bohr frequency: $\omega = -\omega_{fi}$. In this case, the dominant probability amplitude is for the transition from an upper state to a lower state, which we call **emission** [Fig. 14.2(b)]. The system emits energy to the external perturbation. This emission is caused by the applied field, so it is referred to as **stimulated emission**.

Only one of these two terms plays a role in any particular experiment, so we needn't worry about both together. For now, we consider the absorption case only (we just need to change the sign preceding ω if we change to the emission case). The probability of measuring the system in the final state is

$$\mathcal{P}_{i \rightarrow f}(t) = \frac{|V_{fi}|^2}{\hbar^2} \frac{\sin^2 \frac{\omega_f - \omega}{2} t}{\left(\frac{\omega_f - \omega}{2}\right)^2}, \quad (14.29)$$

where we have adopted a shorthand notation for the matrix element of the perturbation:

$$V_{fi} = \langle f | V | i \rangle. \quad (14.30)$$

It is useful to look at this result both as a function of time and as a function of frequency. As a function of time, there is an oscillatory dependence as shown in Fig. 14.3, with a period of $2\pi/(\omega_{fi} - \omega)$. We saw similar results in Chapter 3 for the Rabi oscillations in the spin case (Fig. 3.12), with a slightly different oscillation or flopping frequency. The perturbation result in Eq. (14.29) is equal to the Rabi flopping probability in Eq. (3.104) for the case of small perturbations. In practice, this oscillating probability is hard to observe, which is related to the finite lifetime of excited states that we address in Section 14.5.

As a function of frequency, the transition probability is shown in Fig. 14.4 and displays the expected resonance behavior, with a peak at the probability at $\omega = \omega_{fi}$. The similar result for resonant Rabi spin flopping was shown in Fig. 3.11. The peak of the probability in Fig. 14.4 grows as t^2 , so it could become greater than one, which would violate our perturbation approximation. The resonance curve in Fig. 14.4 has a finite width, which implies that the resonance condition $\omega = \omega_{fi}$ is not an exact requirement. Rather, there is a spread of frequencies $\Delta\omega$ that cause appreciable transition probability. The frequency width of the probability plot in Fig. 14.4 is approximately $\Delta\omega = 4\pi/t$, where the time t is the duration of the interaction. If we call this duration Δt , then we have

$$\Delta\omega\Delta t \approx \frac{4\pi}{\Delta t}\Delta t = 4\pi, \quad (14.31)$$

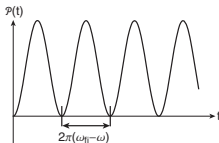


FIGURE 14.3 Oscillations of the transition probability as a function of time.

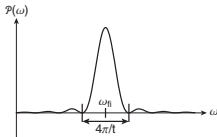


FIGURE 14.4 Probability of excitation as a function of frequency.

which is the Fourier frequency-time uncertainty relation. We can convert this to a Fourier energy-time uncertainty relation by using $\Delta E = \hbar \Delta \omega$ to obtain

$$\begin{aligned} \Delta E \Delta t &\approx \hbar \Delta \omega \Delta t = \hbar 4\pi \\ &\approx \hbar. \end{aligned} \quad (14.32)$$

The neglect of the factor of 4π is consistent with the level of this approximation. This uncertainty relation tells us that the longer we observe a system, the better we can measure the energy.

The resonance peak in Fig. 14.4 resembles a Dirac delta function in the limit that the frequency width $\Delta \omega = 4\pi/t \rightarrow 0$, which implies $t \rightarrow \infty$. Mathematically, the *sinc* function in Eq. (14.29) becomes a Dirac delta function in this limit:

$$\lim_{t \rightarrow \infty} \frac{\sin^2 \frac{\omega_f - \omega}{2} t}{\left(\frac{\omega_f - \omega}{2}\right)^2} = 2\pi t \delta(\omega_f - \omega). \quad (14.33)$$

If we assume this long time limit in Eq. (14.29), we obtain the probability

$$\mathcal{P}_{i \rightarrow f}(t \rightarrow \infty) = \frac{2\pi t}{\hbar^2} |V_{fi}|^2 \delta(\omega_f - \omega). \quad (14.34)$$

This form makes it evident that the probability increases linearly with time. This behavior is more common than the oscillating probability of Fig. 14.3. The linear time dependence seems reasonable because we expect that the more we perturb a system, the more likely it is to undergo a change. The linear time dependence in Eq. (14.34) allows us to define a **transition rate** as the probability per unit time, which we obtain by differentiating the probability:

$$R_{i \rightarrow f} = \frac{d}{dt} \mathcal{P}_{i \rightarrow f}(t), \quad (14.35)$$

with the result:

$$\boxed{R_{i \rightarrow f} = \frac{2\pi}{\hbar^2} |V_{fi}|^2 \delta(\omega_f - \omega)}. \quad (14.36)$$

The delta function in Eq. (14.36) is called the energy conserving delta function—it requires that the quantum of energy causing the transition (e.g., the photon of a laser beam) match the energy difference between the two states. In many practical applications, there is a spread of final energy states

rather than a discrete quantum state. For example, in a solid, the band structure of the electronic energy levels represents a continuous range of allowed states (see Chapter 15). In those cases, the relevant transition rate is a sum over the rates to all accessible states. We assume that these rates are incoherent so that we can add the rates (i.e., probabilities) rather than the amplitudes. We assume that the spread in energies is larger than the width of the *sinc* function that we turned into a delta function but small enough that the rates to all states are the same. Let $g(E)$ be the density of states per unit energy, such that $g(E)dE$ is the number of energy levels between E and $E + dE$. Then the total rate is given by the integral over all the rates:

$$\begin{aligned} R_{i \rightarrow f} &= \int_{E_f - \epsilon}^{E_f + \epsilon} \frac{2\pi}{\hbar^2} |V_{fi}|^2 \delta(\omega_{fi} - \omega) g(E) dE \\ &= \frac{2\pi}{\hbar^2} |V_{fi}|^2 \int_{E_f - \epsilon}^{E_f + \epsilon} \delta(\omega_{fi} - \omega) g(E) \hbar d\omega. \end{aligned} \quad (14.37)$$

The range over which we integrate in Eq. (14.37) is not important because of the Dirac delta function. The resultant transition rate is

$$R_{i \rightarrow f} = \frac{2\pi}{\hbar} |V_{fi}|^2 g(E_f). \quad (14.38)$$

This result is referred to as **Fermi's golden rule**. It is much more practical than Eq. (14.36) because the nonphysical delta function is gone. Fermi's golden rule is general enough that it applies to many types of interactions. In the next section we'll study one particular application.

14.3 ■ ELECTRIC DIPOLE INTERACTION

One of the most important applications of time-dependent perturbation theory is to the interaction between an atom and an electromagnetic field. Studying this problem tells us how lasers and atoms interact, and it also leads to an understanding of the finite lifetime of excited quantum states. The interaction Hamiltonian between an atom and an applied electromagnetic field is the same as we used for the Stark effect in Chapter 10, except that the electric field is now time dependent. We neglect the interaction of the atom with the magnetic component of the electromagnetic field because it is smaller by a factor of the fine structure constant α .

The electric dipole Hamiltonian is

$$H' = -\mathbf{d} \cdot \mathbf{E}. \quad (14.39)$$

The electric field is

$$\begin{aligned} \mathbf{E}(t) &= 2\mathcal{E}_0 \hat{\mathbf{e}} \cos \omega t \\ &= \hat{\mathbf{e}} (\mathcal{E}_0 e^{i\omega t} + \mathcal{E}_0 e^{-i\omega t}), \end{aligned} \quad (14.40)$$

with the same form of the time dependence as in the generic harmonic perturbation example in the previous section. The polarization of the electric field is specified by the unit vector $\hat{\mathbf{e}}$. We ignore the spatial variation of the field because the size of the atom (~ 0.1 nm) is much smaller than the wavelength of visible light (~ 500 nm), which is the most common case. This means that at any given instant, the

whole atom sees the same electric field. This assumption is the **electric dipole approximation**. We use the previous harmonic perturbation calculation and identify the perturbation in Eq. (14.27) as:

$$V = -\mathbf{d} \cdot \hat{\mathbf{e}} \mathcal{E}_0. \quad (14.41)$$

The atom's electric dipole moment is

$$\mathbf{d} = -e\mathbf{r}, \quad (14.42)$$

resulting in the perturbation

$$V = e\mathcal{E}_0 \hat{\mathbf{e}} \cdot \mathbf{r}. \quad (14.43)$$

Application of Fermi's golden rule in the form of Eq. (14.36) yields the transition rate:

$$R_{i \rightarrow f} = \frac{2\pi}{\hbar^2} |\langle f | e\mathcal{E}_0 \hat{\mathbf{e}} \cdot \mathbf{r} | i \rangle|^2 \delta(\omega_f - \omega). \quad (14.44)$$

Only the $\mathbf{r}$ term in the matrix element depends on the atomic states, so we simplify the rate to

$$R_{i \rightarrow f} = \frac{2\pi e^2 \mathcal{E}_0^2}{\hbar^2} |\hat{\mathbf{e}} \cdot \langle f | \mathbf{r} | i \rangle|^2 \delta(\omega_f - \omega). \quad (14.45)$$

The delta function in Eq. (14.45) is not physical, so we must see how to apply this transition rate expression to a real situation. The two most common situations are depicted in Fig. 14.5: (a) the perturbing field is not a single frequency and is not coherent, in which case we sum over transition rates caused by the spread of frequencies; or (b) the quantum energy states are continuous, in which case we sum over transition rates to a spread of energy states, as we did in the last section. The first case is necessary when using a broadband light source excites a discrete atomic transition. The second case is necessary when using a monochromatic laser to excite a system to a spread of excited states, or even to a single excited state that is broadened by its finite lifetime. We start with the first case because it allows us to study the interaction between blackbody radiation and atoms, which Einstein used to model the broadening of atomic states. Once we know how the atomic states are broadened, we'll use that knowledge to study the second case of single frequency excitation.

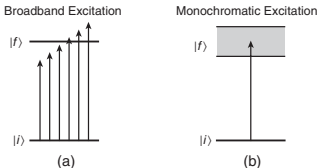


FIGURE 14.5 (a) Broadband excitation to a discrete level, and (b) monochromatic excitation to a broadened level.

14.3.1 ■ Einstein Model: Broadband Excitation

The Einstein model assumes a gas of two-level atoms in thermal equilibrium with blackbody radiation at temperature T . The atoms are considered to have discrete energy levels and the blackbody radiation is modeled as a broadband incoherent electromagnetic field. The goal is to reduce Eq. (14.45) to a simple form for the transition rate involving the atom properties and the field properties.

The electromagnetic field of the blackbody radiation has an energy density per unit volume given by

$$u = \frac{\epsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} B^2. \quad (14.46)$$

The energy density in the electric and magnetic fields is the same. Substituting Eq. (14.40) into Eq. (14.46) gives

$$u = \epsilon_0 \mathcal{E}^2 = 4\epsilon_0 \mathcal{E}_0^2 \cos^2(\omega t). \quad (14.47)$$

The time-average over one cycle gives a factor of 1/2, resulting in

$$u_{rms} = 2\epsilon_0 \mathcal{E}_0^2. \quad (14.48)$$

For broadband radiation, the energy density in the electromagnetic field is

$$u_{rms} = \rho(\omega) d\omega, \quad (14.49)$$

where $\rho(\omega)$ is the field energy per unit volume per unit angular frequency interval. Combining Eqs. (14.48) and (14.49) gives

$$\mathcal{E}_0^2 = \frac{\rho(\omega)}{2\epsilon_0} d\omega, \quad (14.50)$$

which we substitute into Eq. (14.45) to obtain the transition rate. We integrate over all the transition rates due to each frequency component because the blackbody light is incoherent, which gives

$$\begin{aligned} R_{i \rightarrow f} &= \frac{\pi e^2}{\epsilon_0 \hbar^2} |\hat{\mathbf{e}} \cdot \langle f | \mathbf{r} | i \rangle|^2 \int_0^\infty \rho(\omega) \delta(\omega_{fi} - \omega) d\omega \\ &= \frac{\pi e^2}{\epsilon_0 \hbar^2} \rho(\omega_{fi}) |\hat{\mathbf{e}} \cdot \langle f | \mathbf{r} | i \rangle|^2. \end{aligned} \quad (14.51)$$

Inside the black box containing the blackbody radiation and the atoms, the radiation is isotropic and the polarization vector is random, so we average Eq. (14.51) over all possible directions of the polarization vector $\hat{\mathbf{e}}$. To do this average, let θ be the angle between $\hat{\mathbf{e}}$ and $\mathbf{r}$. The three-dimensional spatial average of $|\hat{\mathbf{e}} \cdot \mathbf{r}|^2$ is

$$\begin{aligned} \langle |\hat{\mathbf{e}} \cdot \mathbf{r}|^2 \rangle &= \frac{1}{4\pi} \int |\hat{\mathbf{e}} \cdot \mathbf{r}|^2 d\Omega \\ &= \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi \cos^2 \theta \sin \theta d\theta d\phi \\ &= \frac{1}{4\pi} 2\pi \left[-\frac{1}{3} \cos^3 \theta \right]_0^\pi \\ &= \frac{1}{3}. \end{aligned} \quad (14.52)$$

Thus we get for the transition rate:

$$R_{i \rightarrow f} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \rho(\omega_{fi}) |\langle f | \mathbf{r} | i \rangle|^2. \quad (14.53)$$

Einstein grouped the atomic factors into the now-famous **Einstein B coefficient**

$$B_{if} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} |\langle f | \mathbf{r} | i \rangle|^2 \quad (14.54)$$

and wrote the transition rate as

$$R_{i \rightarrow f} = B_{if} \rho(\omega_{fi}). \quad (14.55)$$

The Einstein *B* coefficient is the same if we swap initial and final states, as is the electromagnetic energy density $\rho(\omega_{fi})$ evaluated at the transition frequency, so the rates of emission and absorption are the same.

In the Einstein model, a collection of atoms is in thermal equilibrium with blackbody radiation at a temperature T . The atoms are treated as having two states $|1\rangle$ and $|2\rangle$ with energies E_1 and E_2 , respectively. The radiation induces transitions from $|1\rangle$ to $|2\rangle$ —absorption, and from $|2\rangle$ to $|1\rangle$ —stimulated emission, as depicted in Fig. 14.6. Because the absorption and stimulated emission rates are the same, the populations of the two levels would be the same if there were no other processes. But the Boltzmann thermal distribution law tells us that the populations of levels decrease as the energy level increases. Einstein argued that there must be a third process—**spontaneous emission**—connecting the two levels in order for thermal equilibrium to be maintained. This is called the principle of **detailed balance**. Spontaneous emission occurs spontaneously, independent of the applied field, and therefore it causes the excited state $|2\rangle$ to decay to the ground state $|1\rangle$ even when no field is present. Because of this spontaneous decay, the excited state $|2\rangle$ has a *finite lifetime*, in contradiction to our previous declaration that all energy eigenstates are stationary states. The transition rate for spontaneous emission was defined by Einstein as A_{21} and is called the **Einstein A coefficient**. The spontaneous emission rate is *independent* of the applied field. The stimulated emission rate *depends* on the applied field according to Eq. (14.55).

Let's now calculate the Einstein A coefficient of spontaneous emission. Assume that there are N_1 atoms in the lower state $|1\rangle$ and N_2 atoms in the upper state $|2\rangle$. The transition rates we have discussed so far are the rates for single atoms, so the rates for a collection or ensemble of atoms are obtained by multiplying the single atom rates by the population of the initial state. For example, the number of atoms per second that absorb photons and change from state $|1\rangle$ to state $|2\rangle$ is the transition rate for

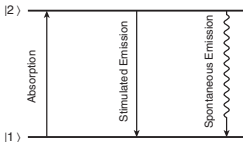


FIGURE 14.6 Einstein model of absorption and emission of photons.

a single atom $B_{12}\rho(\omega_{21})$ times the number of atoms N_1 in state $|1\rangle$: $N_1 B_{12}\rho(\omega_{21})$. This absorption process decreases the number of atoms in state $|1\rangle$, while the two emission processes increase the number. The sum of the three rates yields the **rate equation** for state $|1\rangle$:

$$\frac{dN_1}{dt} = -N_1 B_{12}\rho(\omega_{21}) + N_2 B_{21}\rho(\omega_{21}) + N_2 A_{21}. \quad (14.56)$$

The rate equation for state $|2\rangle$ is similarly

$$\frac{dN_2}{dt} = +N_1 B_{12}\rho(\omega_{21}) - N_2 B_{21}\rho(\omega_{21}) - N_2 A_{21}. \quad (14.57)$$

Note that $dN_1/dt = -dN_2/dt$ because there are only two levels in this model system and all atoms leaving one state end up in the other state.

In steady state, the number of atoms in either state is constant (but not equal to each other), with as many atoms making upward transitions as downward transitions. Hence, the change dN_1/dt equals zero and we can solve Eq. (14.56) for the radiation energy density:

$$\rho(\omega_{21}) = \frac{A_{21}}{B_{12}} \frac{1}{(N_1/N_2)(B_{12}/B_{21}) - 1}. \quad (14.58)$$

The blackbody energy density is determined by the Planck blackbody radiation formula:

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\hbar\omega/k_B T} - 1}, \quad (14.59)$$

which comes from the Boltzmann probability of occupation of the modes of the radiation field. In thermal equilibrium, elementary statistical mechanics tells us that the number of atoms in an energy level E is proportional to the Boltzmann factor $\exp(-E/k_B T)$, where k_B is Boltzmann's constant. Hence, the ratio of level populations is:

$$\frac{N_1}{N_2} = \frac{e^{-E_1/k_B T}}{e^{-E_2/k_B T}} = e^{(E_2-E_1)/k_B T} = e^{\hbar\omega_{21}/k_B T}. \quad (14.60)$$

Combining Eqs. (14.58), (14.59), and (14.60) leads to two conditions:

$$B_{21} = B_{12}, \quad (14.61)$$

which we already knew from Eq. (14.54), and

$$A_{21} = \frac{\hbar\omega_{21}^3}{\pi^2 c^3} B_{21}, \quad (14.62)$$

which relates the Einstein A and B coefficients. Using Eq. (14.54) for the Einstein B coefficient leads us to the spontaneous emission rate:

$$A_{21} = \frac{e^2 \omega_{21}^3}{3\pi \epsilon_0 \hbar c^3} |\langle 2|\mathbf{r}|1\rangle|^2. \quad (14.63)$$

The decay of a state caused by spontaneous emission implies that excited states are not stationary states, as we have assumed all along about quantum energy eigenstates. Rather, excited states have

an inherent finite lifetime due to spontaneous emission. If we have a system of atoms in the excited state, with no electromagnetic fields present, then the rate equation for the upper level is

$$\frac{dN_2}{dt} = -N_2 A_{21}. \quad (14.64)$$

Solving this differential equation yields the time dependence of the upper level population

$$N_2(t) = N_2(0)e^{-A_{21}t} = N_2(0)e^{-t/\tau}. \quad (14.65)$$

The upper level population decays exponentially, as shown in Fig. 14.7, with a lifetime τ given by the inverse of the Einstein A coefficient

$$\tau = \frac{1}{A_{21}}. \quad (14.66)$$

This inherent finite lifetime of the excited state means that there is a fundamental limit to the time we have to observe the system in this state. Therefore, the energy-time uncertainty relation in Eq. (14.32) implies that the finite lifetime of the excited state places a *fundamental limit on how well we can measure the energy*. Hence, the energy of an excited state is uncertain or broadened. The uncertainty in our measurement of the energy difference between the ground and the excited state is

$$\Delta E = \frac{\hbar}{\Delta t} = \frac{\hbar}{\tau} = \hbar A_{21}. \quad (14.67)$$

No matter how precise our measurement apparatus is, we cannot overcome this limitation. The energy uncertainty in Eq. (14.67) is the spread in energy of a state that was depicted in Fig. 14.5(b), which we address in the next section.

Though we now have a way to calculate the spontaneous emission rate, we have not discovered the mechanism that is responsible for the decay of excited states in the absence of a perturbing radiation field. The approach that we have taken here to atom-light interactions is known as the semiclassical method because we have treated the atoms quantum mechanically, but we have treated the light as a classical field. To properly explain spontaneous emission, we must use quantum electrodynamics (QED). Quantum electrodynamics treats the light quantum mechanically as a harmonic oscillator, with the state $|n\rangle$ representing a light field with n photons. The ground state $|0\rangle$ has no photons (i.e., no field excitations), but has an energy $\hbar\omega/2$, just as the ground state of the harmonic oscillator does. This vacuum state energy represents residual energy in the electromagnetic field, which “stimulates”

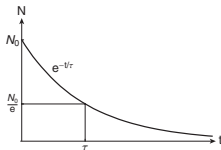


FIGURE 14.7 Exponential time decay of the population of an excited atomic state.

the emission of photons from excited atoms. Hence, spontaneous emission can be considered to be emission that is stimulated by the vacuum. Recent experiments have made this interpretation clear by showing that the spontaneous emission rate can be changed by altering the vacuum, which is possible if you put an atom in a specially sized box, a cavity, that alters the allowed radiation modes at the frequency of interest. The quantum mechanical interaction of atoms and quantized light fields is known as **cavity QED**.

14.3.2 ■ Laser Excitation

We now address the problem of a monochromatic laser exciting an atom with an upper level that is not sharply defined, as depicted in Fig. 14.5(b). We assume that the spread in energy of the upper atomic level is caused by spontaneous emission, though collisions or other environmental factors are also possible causes. Fermi's golden rule in Eq. (14.38) tells us that the transition rate depends on the density of energy states. We found the spread of the final energy state in Eq. (14.67) for the case of spontaneous emission. The functional form of the density of energy states $g(E)$, or equivalently the frequency density, is determined by the Fourier transform of the emitted electromagnetic field. For the exponential time dependence of a spontaneously decaying upper state, this Fourier transform yields a Lorentzian function (Problem 14.5). For a transition between states $|1\rangle$ and $|2\rangle$ with a spontaneous decay rate A_{21} from state $|2\rangle$, the Lorentzian density of states for the upper state is

$$g(E) = \frac{\hbar A_{21}/2\pi}{(E - \hbar\omega_{21})^2 + \left(\frac{\hbar A_{21}}{2}\right)^2}. \quad (14.68)$$

The density of states is normalized to unity, $\int_0^\infty g(E) dE = 1$, because there is only one state at the upper level; it is just spread out by its finite lifetime. Substituting the density of states into Fermi's golden rule in Eq. (14.38) yields the transition rate

$$\begin{aligned} R_{1 \rightarrow 2} &= \frac{2\pi}{\hbar} |V_{21}|^2 g(E_f) \\ &= \frac{2\pi e^2 E_0^2}{\hbar} |\hat{\mathbf{e}} \cdot \langle 2 | \mathbf{r} | 1 \rangle|^2 \frac{\hbar A_{21}/2\pi}{(E - \hbar\omega_{21})^2 + \left(\frac{\hbar A_{21}}{2}\right)^2}. \end{aligned} \quad (14.69)$$

Using the frequency Lorentzian ($f(\omega) d\omega = g(E) dE$)

$$f(\omega) = \frac{A_{21}/2\pi}{(\omega - \omega_{21})^2 + \left(\frac{A_{21}}{2}\right)^2}, \quad (14.70)$$

we express the transition rate as

$$R_{1 \rightarrow 2} = \frac{2\pi e^2 E_0^2}{\hbar^2} |\hat{\mathbf{e}} \cdot \langle 2 | \mathbf{r} | 1 \rangle|^2 f(\omega) \quad (14.71)$$

In terms of the Einstein B coefficient, the transition rate is

$$R_{1 \rightarrow 2} = 6\epsilon_0 \mathcal{E}_0^2 B_{21} f(\omega). \quad (14.72)$$

If we excite the transition with a monochromatic laser with intensity $I = 2c\epsilon_0 \mathcal{E}_0^2$, the transition rate is

$$R_{1 \rightarrow 2} = 3 \frac{I}{c} B_{12} f(\omega). \quad (14.73)$$

This excitation probability rate has the Lorentzian frequency dependence shown in Fig. 14.8, with a full width at half maximum (FWHM) of A_{21} . Once again, we see the resonance behavior of the interaction, such that the laser must be tuned within this frequency window in order to have appreciable probability of inducing excitation of the atom.

Another useful way to quantify the excitation of an atom by a laser is with a quantity known as the **cross section**. To understand why an area is useful in this regard, consider characterizing the efficiency of the interaction as the ratio of *what you get out* to *what you put in*:

$$\text{efficiency} = \frac{\text{output}}{\text{input}}. \quad (14.74)$$

In this case, you put in light and get out excited atoms.

We usually characterize the input laser light in terms of the intensity I , measured in Watts per square meter. However, to simplify matters, let's quantify the light in terms of the number of photons per unit area per unit time. Each photon in the laser beam has an energy $\hbar\omega$, so the number of photons per unit area per unit time is the intensity divided by the energy per photon:

$$\frac{\# \text{ photons}}{\text{area} \cdot \text{time}} = \frac{I}{\hbar\omega}. \quad (14.75)$$

We quantify the output of excited atoms by the transition rate R , which is a probability (i.e., number) per unit time. Thus the efficiency we have defined becomes:

$$\begin{aligned} \text{efficiency} &= \frac{\text{output}}{\text{input}} \\ &= \frac{R_{1 \rightarrow 2}}{\frac{I}{\hbar\omega}} = \frac{\# \text{ per unit time}}{\# \text{ per unit time per unit area}}. \end{aligned} \quad (14.76)$$

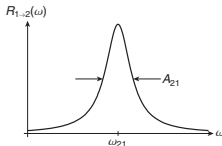


FIGURE 14.8 Lorentzian frequency dependence of the excitation probability.

By dimensional analysis, the efficiency we have defined is really an effective area for excitation. This effective area is what we call the cross section σ . To calculate the cross section, we assume that the laser is on resonance ($\omega = \omega_{21}$) in order to provide the maximum efficiency:

$$\begin{aligned}\sigma &= \frac{R_{1 \rightarrow 2}}{I/\hbar\omega} = \frac{3(I/c)B_{12}f(\omega_{21})}{I/\hbar\omega_{21}} \\ &= \frac{3(I/c)B_{12}(2/\pi A_{21})}{I/\hbar\omega_{21}} \\ &= \frac{6\hbar\omega_{21}}{\pi c} \frac{B_{12}}{A_{21}}.\end{aligned}\quad (14.77)$$

From Eq. (14.62) relating the Einstein rate coefficients, we know the ratio of B_{12} to A_{21} . This allows us to calculate the cross section for on-resonance excitation:

$$\begin{aligned}\sigma &= \frac{6\hbar\omega_{21}}{\pi c} \frac{B_{12}}{A_{21}} = \frac{6\hbar\omega_{21}}{\pi c} \frac{\pi^2 c^3}{\hbar\omega_{21}^3} \\ &= 6\pi \frac{c^2}{\omega_{21}^2} = 6\pi \frac{1}{k_{21}^2} = 6\pi \frac{1}{(2\pi/\lambda_{21})^2},\end{aligned}\quad (14.78)$$

resulting in

$$\boxed{\sigma = 3 \frac{\lambda_{21}^2}{2\pi}}. \quad (14.79)$$

This result is amazingly simple. It says that the atom is effectively the size of the wavelength of light ($\sim 100 - 1000$ nm) when considering its interaction with resonant light! The physical size of the atom (the Bohr radius ~ 0.1 nm) is irrelevant in this case, although it would be more appropriate if we were considering collisions between two atoms. For atom-light interactions, the atom acts as an efficient antenna, despite its small size.

14.4 ■ SELECTION RULES

The Einstein A and B coefficients depend upon the matrix element $\langle f|\hat{\mathbf{s}} \cdot \mathbf{r}|i\rangle$ from the electric dipole interaction between the two states. If this matrix element is zero for some reason, then there is no probability that the transition between the states will occur. There are some general guidelines as to when such matrix elements are expected to be zero, and we call these **selection rules**. Transitions for which the matrix element is zero are therefore not allowed and are called **forbidden transitions**. However, recall that we are working within the electric dipole approximation, which means that we have neglected magnetic dipole, electric quadrupole, and higher multipole interactions. It may happen that a transition is forbidden within the electric dipole approximation but is allowed by a higher-order multipole interaction. The higher-order interactions typically have transition rates that are reduced by an extra order of the fine structure constant α .

The selection rules derive from general properties of the electric dipole matrix elements, not from the details of a specific atom or molecule. To see this, first separate the radial and angular parts of the matrix element:

$$\begin{aligned}
 \langle f | \hat{\mathbf{e}} \cdot \mathbf{r} | i \rangle &= \hat{\mathbf{e}} \cdot \langle f | \mathbf{r} | i \rangle \\
 &= \int_0^\infty r^2 dr \int d\Omega R_{n_f \ell_f}^*(r) Y_{\ell_f}^{m_f}(\theta, \phi) \hat{\mathbf{e}} \cdot \mathbf{r} R_{n_i \ell_i}(r) Y_{\ell_i}^{m_i}(\theta, \phi) \\
 &= \left(\int_0^\infty R_{n_f \ell_f}^*(r) R_{n_i \ell_i}(r) r^3 dr \right) \left(\int Y_{\ell_f}^{m_f}(\theta, \phi) \hat{\mathbf{e}} \cdot \mathbf{r} Y_{\ell_i}^{m_i}(\theta, \phi) d\Omega \right).
 \end{aligned} \tag{14.80}$$

The radial integral *does* depend critically on the details of a specific atom or molecule and is not typically zero. The angular integral, however, depends on the spherical harmonics, which are independent of the details of the central potential (see Section 7.4). The dot product term in Eq. (14.80) is expressed in terms of the angles θ and ϕ between the electric field polarization vector $\hat{\mathbf{e}}$ and the electron position unit vector $\hat{\mathbf{r}}$

$$\hat{\mathbf{e}} \cdot \hat{\mathbf{r}} = \varepsilon_x \sin \theta \cos \phi + \varepsilon_y \sin \theta \sin \phi + \varepsilon_z \cos \theta. \tag{14.81}$$

It is useful to express the trigonometric functions in Eq. (14.81) in terms of the spherical harmonics:

$$\hat{\mathbf{e}} \cdot \hat{\mathbf{r}} = \sqrt{\frac{4\pi}{3}} \left(\varepsilon_z Y_1^0(\theta, \phi) + \frac{-\varepsilon_x + i\varepsilon_y}{\sqrt{2}} Y_1^1(\theta, \phi) + \frac{\varepsilon_x + i\varepsilon_y}{\sqrt{2}} Y_1^{-1}(\theta, \phi) \right). \tag{14.82}$$

Thus, the dot product $\hat{\mathbf{e}} \cdot \hat{\mathbf{r}}$ is proportional to spherical harmonics of order 1. This key point derives from making the electric dipole approximation. Higher-order multipole matrix elements involve higher-order spherical harmonics and hence yield different selection rules.

Using Eq. (14.82), we find that the angular integral of the electric dipole matrix element in Eq. (14.80) becomes three integrals, each of which is an integral of the product of three spherical harmonics:

$$\int Y_{\ell_f}^{m_f}(\theta, \phi) Y_1^m(\theta, \phi) Y_{\ell_i}^{m_i}(\theta, \phi) d\Omega, \tag{14.83}$$

where one spherical harmonic is limited to order 1 by the electric dipole approximation and the index m varies over 1, 0, -1 according to the three terms in Eq. (14.82). You would expect that such an integral over three spherical harmonic functions would be difficult to do. However, we now make use of the Clebsch-Gordan coefficients from Chapter 11 on the addition of angular momenta. We found there that a coupled angular momentum state can be expressed in terms of uncoupled states using the Clebsch-Gordan coefficients. For orbital angular momentum this means that one spherical harmonic can be decomposed into products of pairs of other spherical harmonics. Given this knowledge of Clebsch-Gordan coefficients, we find the angular integral:

$$\int Y_{\ell_f}^{m_f}(\theta, \phi) Y_1^m(\theta, \phi) Y_{\ell_i}^{m_i}(\theta, \phi) d\Omega = \left[\frac{3(2\ell_i + 1)}{4\pi(2\ell_f + 1)} \right]^{\frac{1}{2}} \langle \ell_i 1 m_i m | \ell_f m_f \rangle \langle \ell_i 1 0 0 | \ell_f 0 \rangle. \tag{14.84}$$

The Clebsch-Gordan coefficient $\langle \ell_i 1 m_i m | \ell_f m_f \rangle$ is the key to understanding the selection rules. We know from Chapter 11 that only certain values of the coupled angular momentum quantum

numbers are allowed for a given set of uncoupled angular momentum quantum numbers, and that many entries in the tables of the Clebsch-Gordan coefficients are zero. The Clebsch-Gordan coefficient $\langle \ell_i 1 m_i m | \ell_f m_f \rangle$ characterizes the addition of the uncoupled angular momenta $j_1 = \ell_i$ and $j_2 = 1$ to form the coupled angular momentum $j = \ell_f$. The rules of adding angular momenta limit the values of the coupled angular momentum ℓ_f to:

$$\ell_f = \ell_i + 1, \ell_i, \ell_i - 1. \quad (14.85)$$

The magnetic quantum numbers are also limited by the Clebsch-Gordan coefficient $\langle \ell_i 1 m_i m | \ell_f m_f \rangle$ to:

$$\begin{aligned} m_i + m &= m_f \\ m &= m_f - m_i. \end{aligned} \quad (14.86)$$

A further restriction on the matrix elements comes from a consideration of parity. The integrand in Eq. (14.84) must be even with respect to spatial symmetry inversion for the integral to be nonzero. The spherical harmonics have parity given by $(-1)^\ell$, that is

$$Y_\ell^m(\theta, \phi) = (-1)^\ell Y_\ell^m(\pi - \theta, \phi + \pi). \quad (14.87)$$

For the three cases allowed by Eq. (14.85), the integrand with $\ell_f = \ell_i$ has odd parity ($2\ell_i + 1$), while the integrand with $\ell_f = \ell_i \pm 1$ has even parity ($2\ell_i$ or $2\ell_i + 2$). We conclude that the angular integral with $\ell_f = \ell_i$ is identically zero, and find another rule:

$$\ell_f \neq \ell_i. \quad (14.88)$$

We say that $\ell_f = \ell_i$ is not allowed by parity. Combining the rules in Eqs. (14.85), (14.86), and (14.88), we find the selection rules for *electric dipole transitions*:

$$\boxed{\begin{aligned} \Delta \ell &= \pm 1 \\ \Delta m &= 0, \pm 1 \end{aligned}}. \quad (14.89)$$

These selection rules arise solely from the angular integral in Eq. (14.80) and they reflect the conservation of angular momentum of the system of atom and photon. The photon has an angular momentum or spin of 1. When the atom absorbs or emits a photon, the final atomic state must reflect the change in angular momentum of the electromagnetic field.

Application of the $\Delta \ell = \pm 1$ selection rule to the hydrogen atom limits the possible spontaneous emission transitions to those shown in Fig. 14.9. For example, within the $n = 2$ level, the $2p$ state can decay to the $1s$ state, and does so with a 1.6 ns lifetime (Problem 14.7). However, the $2s$ state cannot decay to the ground state because $\Delta \ell = 0$ is not allowed. The Lamb shift does displace the $2s_{1/2}$ state slightly above the $2p_{1/2}$ state to which it can decay, but the transition rate is very small due to the cube of the Bohr frequency in Eq. (14.63). Hence, the $2s_{1/2}$ state has a very long decay lifetime of $1/7$ sec, which is caused by a two-photon decay mechanism to the ground state.

Application of the $\Delta m = 0, \pm 1$ selection rule to the hydrogen $2p \rightarrow 1s$ transition yields the allowed transitions shown in Fig. 14.10, for the case of emission. In the $\Delta m = +1$ transition, the atom gains one unit of angular momentum projection along the z -axis, so the emitted photon must have one unit of angular momentum projection in the negative z -direction. Such a photon is called

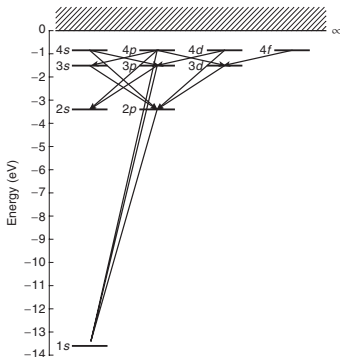


FIGURE 14.9 Decay scheme of hydrogen for allowed electric dipole transitions.

a σ^- polarized photon or a photon with negative **helicity**. The $\Delta m = -1$ transition produces a σ^+ photon with positive helicity. The σ^+ and σ^- photons have a polarization vector that rotates around the z -axis and are also called circularly polarized states. The $\Delta m = 0$ transition produces a photon that has linear polarization along the z -axis, which is referred to as a π polarized photon. For the case of absorption, the Δm values in Fig. 14.10 change sign, but the σ^+ and σ^- labels remain unchanged (Problem 14.13).

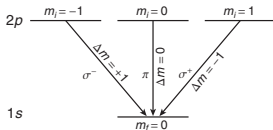


FIGURE 14.10 Hydrogen $2p \rightarrow 1s$ transition.

SUMMARY

In *time-dependent* perturbation theory, we focus on finding the probability that an applied perturbation causes a transition between energy levels of the unperturbed Hamiltonian. In contrast, in *time-independent* perturbation theory, we focus on finding the changes in energy levels caused by the perturbing Hamiltonian (assumed static).

The probability amplitude for a transition from the initial state $|i\rangle$ to the final state $|f\rangle$ subject to the time-dependent perturbation $H'(t)$ is

$$c_f(t) = \frac{1}{i\hbar} \int_0^t \langle f | H'(t') | i \rangle e^{i(E_f - E_i)t'/\hbar} dt'. \quad (14.90)$$

The probability of the transition is

$$\mathcal{P}_{i \rightarrow f}(t) = \frac{1}{\hbar^2} \left| \int_0^t \langle f | H'(t') | i \rangle e^{i(E_f - E_i)t'/\hbar} dt' \right|^2. \quad (14.91)$$

For harmonic perturbation at frequency ω , the transition probability for long times grows linearly with time and we define the transition rate

$$R_{i \rightarrow f} = \frac{2\pi}{\hbar^2} |V_{fi}|^2 \delta(\omega_{fi} - \omega). \quad (14.92)$$

For an electric dipole interaction, the transition rate is

$$R_{i \rightarrow f} = B_{if} \rho(\omega_{fi}) \quad (14.93)$$

if the excitation source is broadband where $\rho(\omega_{fi})$ is the energy density and B_{if} is the Einstein B coefficient

$$B_{if} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} |\langle f | \mathbf{r} | i \rangle|^2. \quad (14.94)$$

If the transition is excited with a monochromatic source with intensity $I = 2c\epsilon_0 \mathcal{E}_0^2$, the transition rate is

$$R_{1 \rightarrow 2} = 3 \frac{I}{c} B_{12} f(\omega), \quad (14.95)$$

where $f(\omega)$ is the frequency response function of the atom.

An excited state in an atom has a finite lifetime due to spontaneous emission. The lifetime is the inverse of the Einstein A coefficient

$$A_{21} = \frac{e^2 \omega_{21}^3}{3\pi \epsilon_0 \hbar c} |\langle 2 | \mathbf{r} | 1 \rangle|^2. \quad (14.96)$$

Electric dipole transitions have the selection rules

$$\begin{aligned} \Delta \ell &= \pm 1 \\ \Delta m &= 0, \pm 1. \end{aligned} \quad (14.97)$$

PROBLEMS

- 14.1** Use the perturbation series expansion of the coefficient c_n given by Eq. (14.11) in the differential equation (14.10) and verify the zeroth-order and first-order equations (14.12) and (14.13). You may wish to use the λ notation of Chapter 10 to keep track of orders.
- 14.2** A particle of mass m is initially in the ground state (E_1) of an infinite square well of width L . Starting at time $t = 0$, the system is subject to the perturbation

$$H'(t) = V_0 x^2 e^{-t/\tau},$$

where V_0 and τ are constants. Find the probability that the energy after time T is measured to be E_2 . Calculate the probability in the limit $T \rightarrow \infty$.

- 14.3** A particle of mass m is initially in the ground state (E_1) of an infinite square well of width L . Starting at time $t = 0$, the system is subject to the perturbation

$$H'(t) = V_0 x e^{-\alpha t^2},$$

where V_0 and α are constants.

- a)** Find the probability that the energy is measured to be E_2 in the limit $t \rightarrow \infty$.
b) Find the probability that the energy is measured to be E_3 in the limit $t \rightarrow \infty$.
- 14.4** A particle of mass m is initially in the ground state (E_1) of an infinite square well of width L . From $t = 0$ to $t = T$, the potential is perturbed so that it becomes

$$V(x) = \begin{cases} V_0, & 0 < x < L/2 \\ 0, & L/2 < x < L \\ \infty, & \text{elsewhere,} \end{cases}$$

where $V_0 \ll E_1$. Find the probability that the energy after time T is measured to be E_2 .

- 14.5** Spontaneous emission causes the population of an atom to decay with the form $e^{-t/\tau}$. The radiated electromagnetic power exhibits this same time dependence, but the field has the form $e^{-t/2\tau}$ because the power is proportional to the field squared. Calculate the Fourier transform of the emitted field and take its complex square to find the frequency spectrum of the radiated power in spontaneous emission. Convert this frequency spectrum to an energy spectrum and normalize it to unity to verify the energy density of states in Eq. (14.68).
- 14.6** A hydrogen atom in its ground state is subject to an applied electric field

$$\mathbf{E} = \mathcal{E}_0(\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}})e^{-t/\tau}.$$

Find the probabilities that after a long time the atom is found to be in each of the four $n = 2$ states.

- 14.7** Calculate the lifetime (in seconds) of each of the four $n = 2$ states of hydrogen ($|n\ell m\rangle$). The lifetime is the inverse of the spontaneous emission rate (Einstein A coefficient).
- 14.8** A particle in a square well potential (with walls at $x = 0$ and $x = L$; that is, $V(x) = 0$ for $0 < x < L$; $V(x) = \infty$ otherwise) starts out in the ground state

$$|\psi(t = 0)\rangle = |1\rangle,$$

where $|n\rangle$ are the normalized eigenstates of the unperturbed Hamiltonian. Starting at $t = 0$, a time-dependent perturbation is applied given by

$$H'(x, t) = V_0 \sin \frac{\pi x}{L} e^{-\gamma t}.$$

- a) Calculate the probability for the particle to make a transition to an excited state $|n\rangle$ ($n \neq 1$) after a long time. Define “long time.”
 b) Are there any selection rules for this transition? If so, what are they?

- 14.9** A particle in the harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$ starts out in the ground state

$$|\psi(t=0)\rangle = |0\rangle,$$

where $|n\rangle$ are the normalized eigenstates of the Hamiltonian. Starting at $t = 0$, a time-dependent perturbation is applied given by

$$H'(x, t) = Ax^3 e^{-\gamma t}.$$

- a) Calculate the probability for the particle to make a transition to an excited state $|n\rangle$ ($n \neq 0$) after a long time. Define “long time.”
 b) Are there any selection rules for this transition? If so, what are they?
- 14.10** Consider two possible types of electric dipole transitions: a $p \rightarrow s$ transition and a $p \rightarrow d$ transition. In each case, choose one *allowed* set of m quantum numbers and explicitly perform the angular integral in Eq. (14.84), then use the Clebsch-Gordan Table 11.5 to confirm your result.
- 14.11** Use the result in Eq. (14.84) and the Clebsch-Gordan Table 11.5 to identify each possible electric dipole transition from an initial p state. Identify the particular Clebsch-Gordan coefficient in Table 11.5 that represents the parity rule $\ell_f \neq \ell_i$.
- 14.12** A hydrogen atom starts in the state $n = 4$, $l = 3$, $m_l = 3$, where we ignore the spin. What possible states will the atom go through as it decays to the ground state? What are the polarizations of the photons that are emitted?
- 14.13** Draw a transition diagram like Fig. 14.10, but for absorption from $1s \rightarrow 2p$ rather than emission. Explain why the Δm values change sign but the σ^+ and σ^- labels remain unchanged.
- 14.14** A particle of mass m and charge q is confined in a one-dimensional harmonic oscillator potential of natural frequency ω .
- What are the selection rules governing spontaneous emission from excited states?
 - Which states can decay directly to the ground state?
 - Find the spontaneous emission rate from the first excited state to the ground state.
 - Calculate the lifetime of the first excited state for an electron bound in a potential with $\omega = 10^{15}$ rad/s.

RESOURCES

Further Reading

More details on atom-light interactions can be found in:

- A. Corney, *Atomic and Laser Spectroscopy*, Oxford: Clarendon Press, 1977.
 C. J. Foot, *Atomic Physics*, Oxford: Oxford University Press, 2005.
 M. Fox, *Quantum Optics*, Oxford: Oxford University Press, 2006.
 R. Loudon, *Quantum Theory of Light*, Oxford: Oxford University Press, 2000.